A Triangle Through a Looking Glass: Reflections, Circles, and Miquel's Point

N-20220708

One of the pleasures of triangle geometry is that a construction which looks accidental from one viewpoint becomes unavoidable from another. Reflecting the orthocenter over a side lands it on the circumcircle; an angle bisector hides an excenter inside a second circle; and three apparently unrelated circles can be forced through one point. The same two moves keep appearing: change the picture without changing an angle, then recognize four points on a circle.[1]

§1.1 A reflection that knows the circumcircle

Let H be the orthocenter of a nondegenerate triangle ABC. Reflect H across the line BC to obtain X. Why should X lie on the circumcircle? Two perpendicular altitudes already determine the required angle.

For any point Q off the three sidelines, write $\angle BQC$ for the directed angle between the lines QB and QC, taken modulo 180°. Reflection in BC reverses the direction of angles, and hence

$$\measuredangle BXC \equiv -\measuredangle BHC \pmod{180^\circ}.$$

But $BH \perp AC$ and $CH \perp AB$, so

$$\measuredangle BHC \equiv -\measuredangle BAC \pmod{180^\circ}.$$

It follows that

$$\boxed{\measuredangle BXC \equiv \measuredangle BAC \pmod{180^\circ}},$$

which is exactly the criterion for A, B, C, X to be concyclic. Directed angles make the argument independent of whether the triangle is acute or obtuse.

There is a second reflection with an even more surprising outcome. Let M be the midpoint of BC, and reflect H through M, obtaining Y. Then

$$\overrightarrow{BY} = \overrightarrow{HC}, \qquad \overrightarrow{CY} = \overrightarrow{HB}.$$

Since $CH \perp AB$ and $BH \perp AC$, we get

$$BY \perp BA, \qquad CY \perp CA.$$

Thus both $\angle ABY$ and $\angle ACY$ are right angles. By Thales' theorem, B and C lie on the circle whose diameter is AY. This circle is the circumcircle of ABC. In particular, Y is the point on the circumcircle opposite A across a diameter.

[1] The standard reflection and Miquel configurations are developed in R. A. Johnson, *Advanced Euclidean Geometry*, and Evan Chen, *Euclidean Geometry in Mathematical Olympiads*.

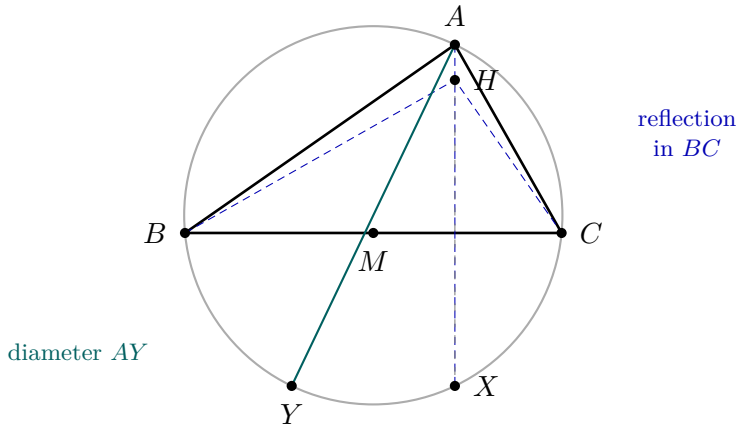


Figure 1.1: Two reflections of the same orthocenter give two distinct points on the circumcircle. The line AY is a diameter.

§1.2 The midpoint of an arc hides an excenter

Let I be the incenter, and let the internal angle bisector AI meet the circumcircle again at L. The equal inscribed angles at A show that L is the midpoint of the arc BC not containing A, so $LB = LC$.

A slightly more useful identity is

$$\boxed{LI = LB = LC.}$$

For example, in triangle BIL,

$$\angle IBL = \frac{A+B}{2} = 90^\circ - \frac{C}{2}.$$

Also, since $\angle BIC = 90^\circ + A/2$ and A, I, L are collinear with I between A and L, the angle $\angle BIL$ equals $90^\circ - C/2$. Hence $LI = LB$; symmetry gives $LI = LC$.

Now reflect I *through the point* L, not across a line, and call the image I_A. Since $L = (I + I_A)/2$, a simple vector identity gives

$$(B - I) \cdot (B - I_A) = |B - L|^2 - |I - L|^2 = 0.$$

The same holds with C in place of B. Thus $BI_A \perp BI$ and $CI_A \perp CI$. The lines BI and CI are internal bisectors, so BI_A and CI_A are their external bisectors. Their intersection I_A is precisely the A-excenter.

The right angles also show that B and C lie on the circle with diameter II_A, whose center is L. A small reflection has simultaneously produced the excenter and a new cyclic quadrilateral. This is often more efficient than introducing excentral distances from scratch.

§1.3 Directed Angles and Invariance

A circle test may be written as

$$\boxed{A, B, C, D \text{ concyclic} \iff \measuredangle ABC \equiv \measuredangle ADC \pmod{180^\circ},}$$

provided the four points are distinct. If one point crosses a side extension, an ordinary acute-angle diagram may reverse an angle or introduce a supplement. Working modulo 180° automatically handles both changes.

The algebra is particularly convenient because adjacent directed angles add:

$$\measuredangle(\ell_1, \ell_2) + \measuredangle(\ell_2, \ell_3) \equiv \measuredangle(\ell_1, \ell_3) \pmod{180^\circ}.$$

But one must not divide arbitrary congruences by two. From $2\alpha \equiv 2\beta \pmod{180^\circ}$ one can infer only $\alpha \equiv \beta \pmod{90^\circ}$. The choice of modulus carries real mathematical information.

The cyclicity criterion is a useful model of a *local test*: a single equality of directions detects a global circle through four points. Miquel's theorem follows by applying this test three times and noticing that the final equality is forced by the first two.

§1.4 Three circles that cannot help meeting

Choose points $D \in BC$, $E \in CA$, $F \in AB$, on the full lines if necessary. Consider the circumcircles

$$\omega_A = (AEF), \qquad \omega_B = (BFD), \qquad \omega_C = (CDE).$$

Assume for the moment that the points and circles are nondegenerate and that ω_A, ω_B have a second intersection P distinct from F. Miquel's theorem asserts that ω_C passes through that same P.

The proof is short enough to see the entire mechanism. Because A, E, F, P are concyclic,

$$\measuredangle EPF \equiv \measuredangle EAF.$$

Because B, F, D, P are concyclic,

$$\measuredangle FPD \equiv \measuredangle FBD.$$

Add the equations and use A, E, C collinear, A, F, B collinear and B, D, C collinear:

$$\begin{aligned} \measuredangle EPD &\equiv \measuredangle EAF + \measuredangle FBD \\ &\equiv \measuredangle(AC, AB) + \measuredangle(AB, BC) \\ &\equiv \measuredangle(AC, BC) \equiv \measuredangle ECD \pmod{180^\circ}. \end{aligned}$$

Therefore C, D, E, P are concyclic, as required. The nondegenerate case proves the theorem; coincident and tangent limiting configurations can be treated by continuity where the limiting circles remain defined.

The interesting point is that the third circle was never independently solved for. Two cyclicity constraints already leave no freedom in the last directed angle. Miquel's point is where this compatibility becomes visible.

§1.5 Inversion of the Three Circles

There is a delightful second way to look at the same configuration *after* finding P. Invert the plane about P. Every circle through P becomes a line, so the three Miquel circles become three straight lines. Their pairwise intersections are exactly the images of F, D, E.

The transformation can be verified without a picture. Put P at the origin and use complex coordinates. A circle through the origin with center $c \neq 0$ satisfies

$$|z - c|^2 = |c|^2 \quad \Longleftrightarrow \quad |z|^2 - 2\,\mathrm{Re}(\overline{c}z) = 0.$$

Under unit inversion $w = z/|z|^2$, the equation becomes

$$\boxed{2\,\mathrm{Re}(\overline{c}w) = 1,}$$

the equation of a straight line. Three circles sharing P therefore become three lines forming an ordinary triangle, with D', E', F' at its vertices.

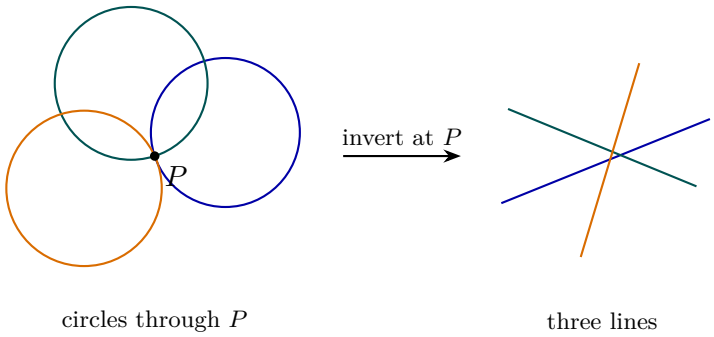


Figure 1.2: A structural view of Miquel's theorem: inversion at the common point straightens all three circles. The diagram illustrates the transformation rather than specific labelled coordinates.

Notice what inversion does *not* do: it is not a projective transformation preserving arbitrary lengths and ratios. It is a conformal transformation away from its center, carrying generalized circles to generalized circles. This is the natural geometry of Miquel configurations.

§1.6 A circle has an algebraic test, too

For distinct complex numbers z_1, z_2, z_3, z_4, define their cross ratio by

$$[z_1, z_2; z_3, z_4] = \frac{(z_3 - z_1)(z_4 - z_2)}{(z_3 - z_2)(z_4 - z_1)}.$$

The four points lie on a generalized circle (a circle or a line) precisely when this quantity is real. To see why, apply the Möbius transformation

$$T(z) = \frac{(z - z_1)(z_2 - z_4)}{(z - z_4)(z_2 - z_1)}.$$

It sends z_1, z_2, z_4 to $0, 1, \infty$. The unique generalized circle through these three images is the real line together with infinity. The fourth point lies on the original generalized circle exactly when its image is real. Up to a real fractional-linear rearrangement, that image is the cross ratio above.

This criterion explains the shared algebraic structure behind directed-angle chasing and inversion. An equality of inscribed angles makes four points concyclic; reality of a cross ratio detects the same property in complex coordinates; a Möbius map can turn the circle into a line where the statement is immediate.

For ordinary triangle problems, a two-line directed-angle proof is usually the most efficient. The higher-level interpretation becomes worthwhile when circles proliferate: instead of chasing every arc, we can choose a conformal coordinate system that turns the relevant circles into straight lines.